VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Vishwakarma Application ID

VK26-IITI-HT-0050

Dermasense: a laser-written moisture strip that reads a wound dressing's own 'change line'

Team: Soham Prabhakar Kadu · Bhakti Ajay Kadam

Track: Health Tech: Emergency Response & Remote Monitoring

Core innovation: two laser-carbonised electrode zones, at the pad centre and on the dressing maker's change line, let a clip-on reader say OK, CHANGE or CHECK without lifting the dressing.

Evidence key on every slide: • completed & verified • designed, physical validation pending • future scope

Problem Context

PUBLISHED EVIDENCE · CITED

44.9 %	≈ 2 cm	1 point	Uneven
OF 588 CHANGES MADE AT OPTIMAL MOISTURE[1]	MAKER'S 'CHANGE NOW' LINE FROM PAD EDGE[2]	WHERE TODAY'S MOISTURE SENSOR READS[1]	IN-VITRO MOISTURE MAP, 8 POINTS[3]

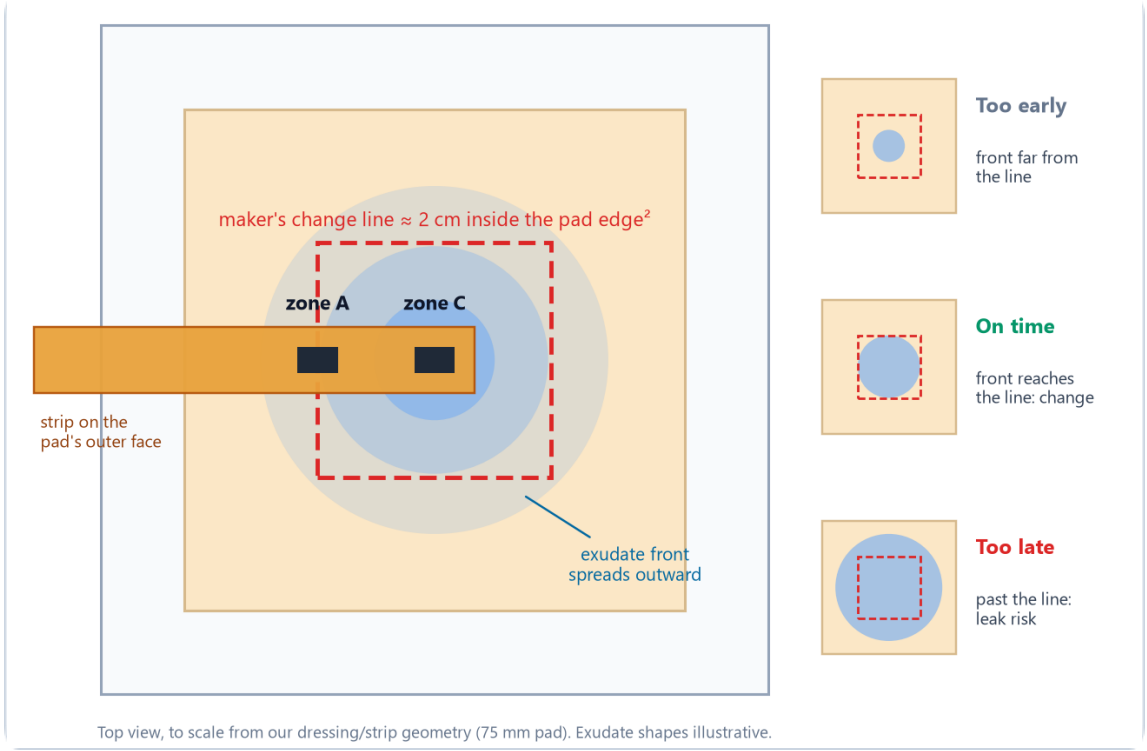


Figure: the maker's change line (≈ 2 cm inside the pad edge[2]) is invisible under a cover. To scale.

Target Beneficiary & Problem Statement

- **Target user:** nurses and home carers looking after exuding chronic wounds (diabetic foot, venous leg and pressure ulcers) under foam dressings.
- **Operational pain point:** the 'change now' signal is visual. Under bandages, socks or compression it is hidden, so dressings are changed by the clock, by lifting the cover, or after a leak.
- **Problem statement:** carers need a reliable, low-cost signal that exudate has reached the maker's change line, without lifting the dressing.
- **Public search disclosure:** WoundSense reads moisture at one point[1]; a lab study mapped 8 points[3]; diaper wetness strips with clip-on readers exist[4]; smart bandages use custom electronics[5]. In our search, none reads a dressing's own change line.
- **Why now:** some makers already define 'full' as a distance from the pad edge[2], and conductive carbon can be laser-written onto polyimide in one pass[7]: a sensor can follow the maker's own rule at low cost.

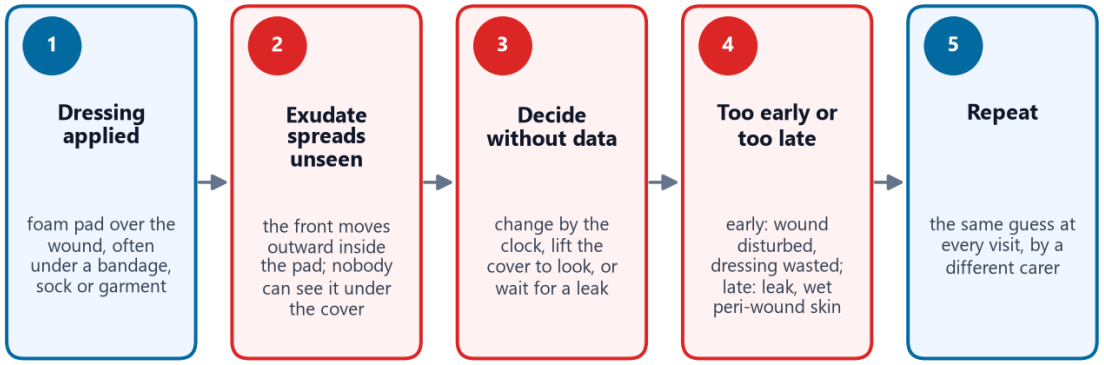
Project Overview: Dermasense · Team ID VK26-IITI-HT-0050 · Track: Health Tech · Core innovation: a laser-written two-zone strip that tells carers when a dressing has reached its own change line.

User Journey: Before State

CURRENT STATE · CITED

44.9 %	0.1–1.3 g/cm²·day	Hidden	Lift to look
CHANGES WHILE MOISTURE WAS STILL OPTIMAL[1]	REPORTED EXUDATE RATES, 2 METHODS[6]	EXUDATE FRONT UNDER BANDAGE OR GARMENT	THE ONLY WAY TO CHECK A COVERED PAD

Today: the dressing is checked by schedule or by looking



- 44.9 % of 588 dressing changes were made while moisture was still optimal (WoundSense study)[1]
- Some makers' rule is visual: change when exudate nears ~2 cm from the pad edge[2]
- Moisture inside a dressing is uneven, so one point is not enough[3]

Quantified figures are published evidence, cited on the References slide

Figure: current pain-point map. Numbers are cited; steps follow dressing instructions.

Current Pain Point Mapping

- **Time loss:** each check means unwrapping the limb or lifting the border, then re-securing it. Minutes per check will be measured in our nurse interviews (not yet done).
- **Material waste:** in the WoundSense study 44.9 % of 588 changes happened while moisture was still optimal[1]: dressings with capacity left were discarded and the wound disturbed.
- **Safety hazard:** a late change lets exudate reach the border or leak: wet peri-wound skin (maceration risk) and soiled clothing.
- **Why a schedule fails:** reported exudate production ranged from 0.10 to 1.3 g/cm² per day across patients and two methods[6]; a fixed interval cannot follow that.
- **Root cause:** the decision signal, the exudate front against the maker's line[2], is visual, and the dressing is covered.

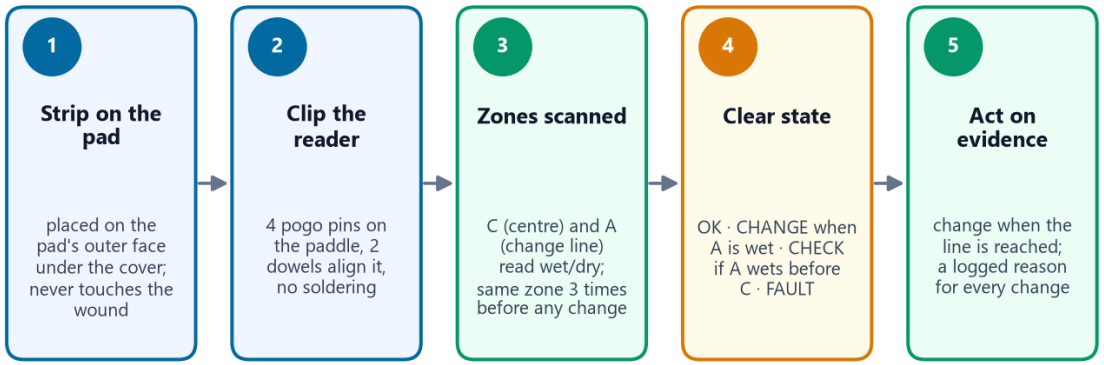
Quantified Inefficiency: only published figures are used; we have not yet measured nurse time or dressing cost per patient, and we do not estimate them here.

User Journey: After State (Intended)

INTENDED WORKFLOW · NOT CLINICALLY VALIDATED

2 zones	4 states	±10 %	0
CENTRE C + MAKER'S CHANGE LINE A	OK · CHANGE · CHECK · FAULT	TARGET ALERT ACCURACY (BENCH, E4)	CONTACT WITH THE WOUND BED (BY DESIGN)

Intended: the strip reads the maker's change line for the carer



- Target (bench, E4): alert within ±10 % of the fluid volume that visibly reaches the change line
- Target: no change of today's dressing routine; disposable = film + carbon only
- Not claimed: fewer changes or faster healing; that needs clinical studies later

INTENDED WORKFLOW: designed and bench-testable, NOT clinically validated

Figure: intended workflow, designed and bench-testable, not clinically validated.

Future Solution Mapping & Quantified Efficiency

- **Fits today's routine:** the strip goes on the pad's outer face when the dressing is applied; the reader clips onto its paddle. Nothing else changes.
- **Decision support:** CHANGE when the front reaches the maker's line; CHECK if A wets before C (edge leak or misplaced strip); FAULT for a broken strip or reader error.
- **Efficiency targets (to measure in E4):** alert within ±10 % of the fluid volume at which the front visibly reaches the line; zero lifts of the dressing just to look.
- **Cost target:** ≤ ₹100 per disposable strip: polyimide film + laser time only; the reader is reused (estimate from the BOM film price).
- **Who benefits (expected, to verify):** patients: fewer needless removals; carers: a logged reason for each change; clinics: fewer wasted dressings.

Design Principle: the strip never touches the wound bed; only film and carbon are thrown away; every electronic part lives in the reusable reader.

System Architecture & Specs

DESIGNED · PHYSICAL VALIDATION PENDING

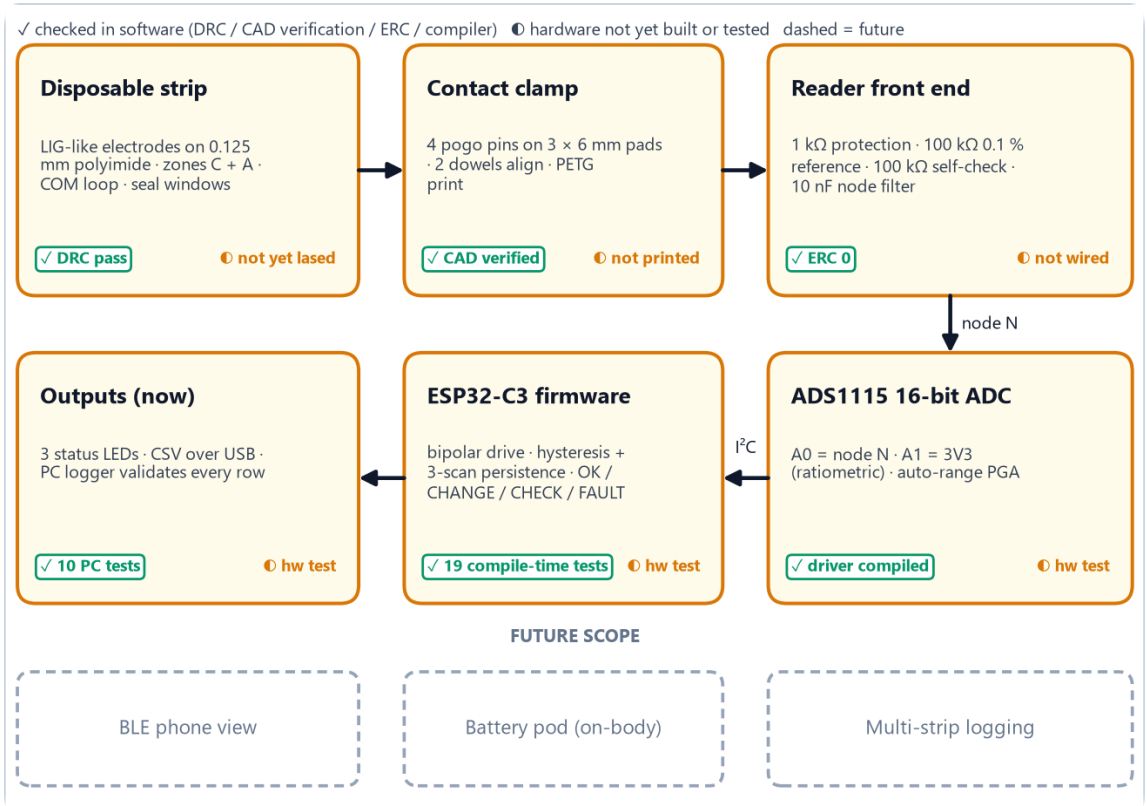


Figure: block diagram with the software-verified and hardware-pending status of each block.

Engineering Target Specifications & Subsystems

Parameter	Target / design value	Selected hardware / basis
Sensing zones	2: C (pad centre), A (change line); windows 6 × 3.9 mm	LIG-like electrodes on 0.125 mm polyimide
Electrode line / gap	1.0 / 1.0 mm (provisional)	fixed by E2 line-width and isolation data
Wet / dry decision	< 100 kΩ wet, > 300 kΩ dry, 3 scans	calc: wet 4–15 kΩ, dry ≥ 1 MΩ
Excitation	bipolar, ≤ 33 µA, idle high-Z	ESP32-C3 GPIO through 1 kΩ
Measurement	16-bit, auto-range, ratiometric	ADS1115 (I²C 0x48), 100 kΩ 0.1 % ref
Self-test	check resistor ±5 % + COM loop	100 kΩ 0.1 %; FAULT state
Contacts	4 pads at 5.08 mm, 2 dowels	P75-B1 pogo pins, PETG clamp
Output	CSV over USB + 3 LEDs (now)	BLE phone view = future scope

Architecture Rationale: only film and carbon are disposable. Bipolar drive keeps the net DC on wet electrodes near zero, and a known resistor checks the reader on every scan.

System Architecture & Specs: Firmware & Data Flow

VERIFIED IN SOFTWARE · HARDWARE PENDING

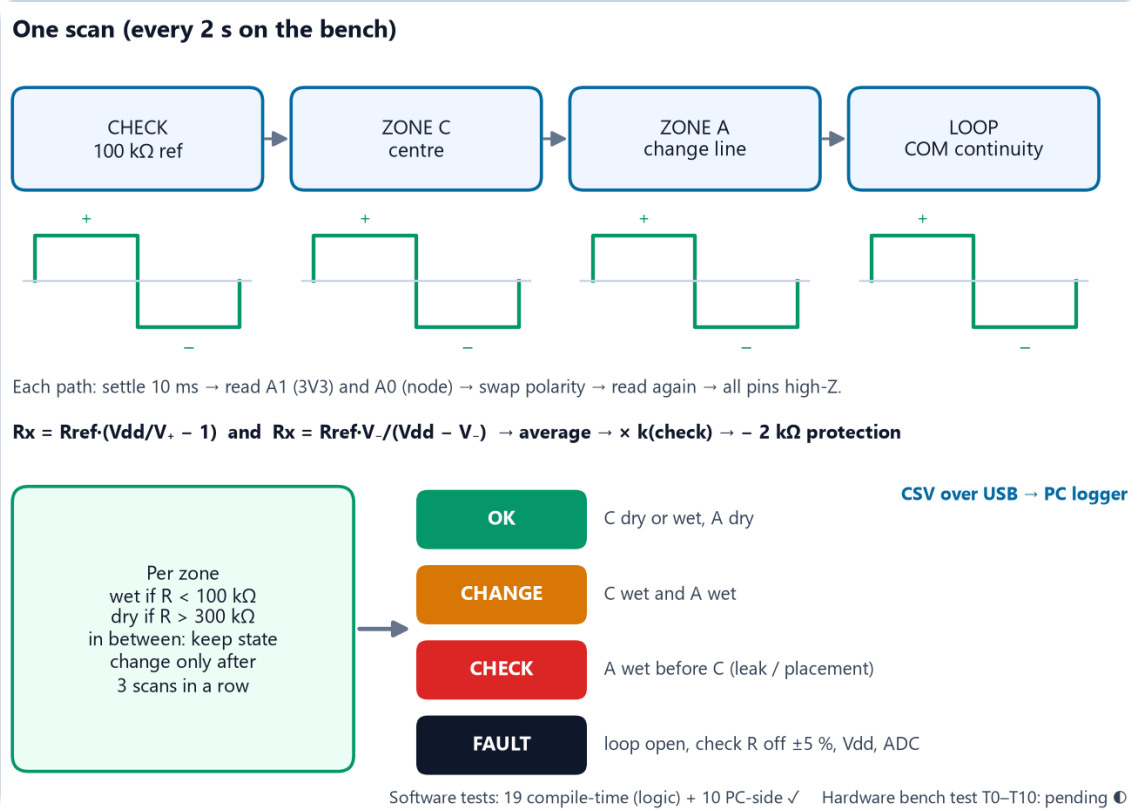


Figure: one reader scan: four bipolar measurements, then hysteresis and the state decision.

Firmware Logic, Signal Measurement & Logging

- **Resistance measurement:** each path is driven both ways through a 100 kΩ 0.1 % reference. Equal + and - phases leave no net DC on wet electrodes.
- **ADC acquisition:** ADS1115 single-shot at 128 SPS, auto-ranged. A1 reads the 3V3 rail, so the maths is ratiometric.
- **Processing:** Rx from both phases, averaged, scaled by the on-board 100 kΩ check (must read within ±5 %, else FAULT), minus the 2 kΩ of protection resistors.
- **Interpretation:** wet below 100 kΩ, dry above 300 kΩ, three scans in a row; LEDs show OK, CHANGE, CHECK (blinking) or FAULT.
- **Logging:** one CSV line per scan over USB. The PC logger checks the header and every row, adds PC time and keeps bad lines apart.
- **Fail-safe behaviour:** between scans every drive pin is high-impedance, so no current flows through the strip; during a FAULT the zone states are frozen, not guessed.

Software vs Hardware: 19 compile-time logic tests and 10 PC-side tests pass; the firmware compiles for ESP32-C3 with 0 warnings. It has not run on a wired board yet (bench test T0–T10 is next).

CAD Design & Mechanical Specs

DESIGNED · CAD VERIFIED · NOT YET BUILT

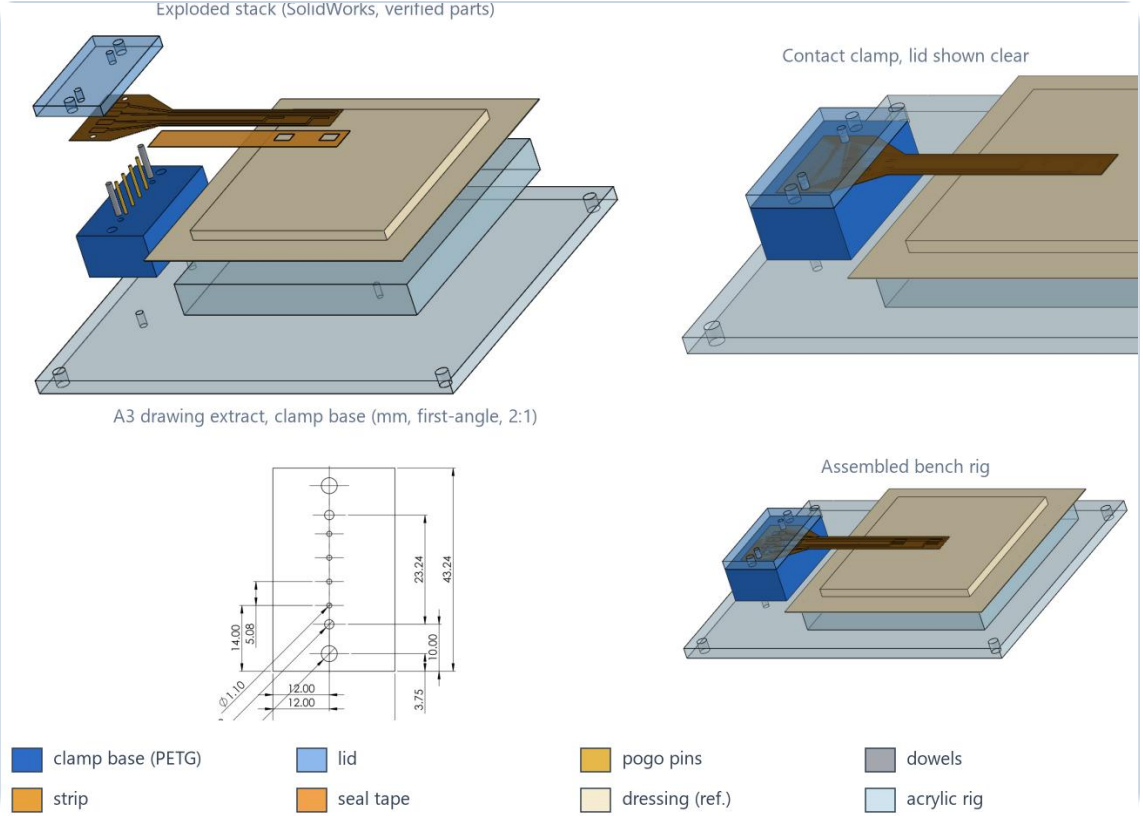


Figure: our SolidWorks 2019 models (9 parts, 1 assembly, 7 A3 drawings, STEP), v0.1 bench rig.

Material Selection & Mechanical Validation

- **Materials:** polyimide 0.125 mm (laser-carbonisable, flexible); PETG clamp (printable, tough); cast acrylic rig (laser-cut); stainless 2 mm dowels; P75-B1 pogo pins.
- **Contact force (hand calculation):** 1.48 mm pin compression → 0.72 N per pin, 2.9 N total; the 4 mm lid deflects 11 µm, so the springs set the force.
- **Alignment:** worst-case tolerance stack 0.49 mm against ±1.35 mm allowed on a 3 mm pad (2.8× margin).
- **Strip loads:** 0.62 % bending strain at a 10 mm radius (polyimide yields near 3 %); a 10 N tug gives 20 MPa at the dowels against 231 MPa tensile.
- **No FEA:** loads are below 3 N, so hand calculations suffice; the contact itself is tested physically (T-CON-01).
- **Independent check:** verify_cad.py re-opens every file: units, volumes, 13 positions, interference, drawings → 99 pass, 0 fail, 5 waiting for E2.

Design Iteration: interference detection found the seal tape trapped under the clamp (9.0 mm³). The seal now starts at x = 23 mm, re-verified with 0 interference (decision D-013).

CAD Design & Mechanical Specs: Sensing Strip & Laser Process

DESIGNED · PHYSICAL VALIDATION PENDING

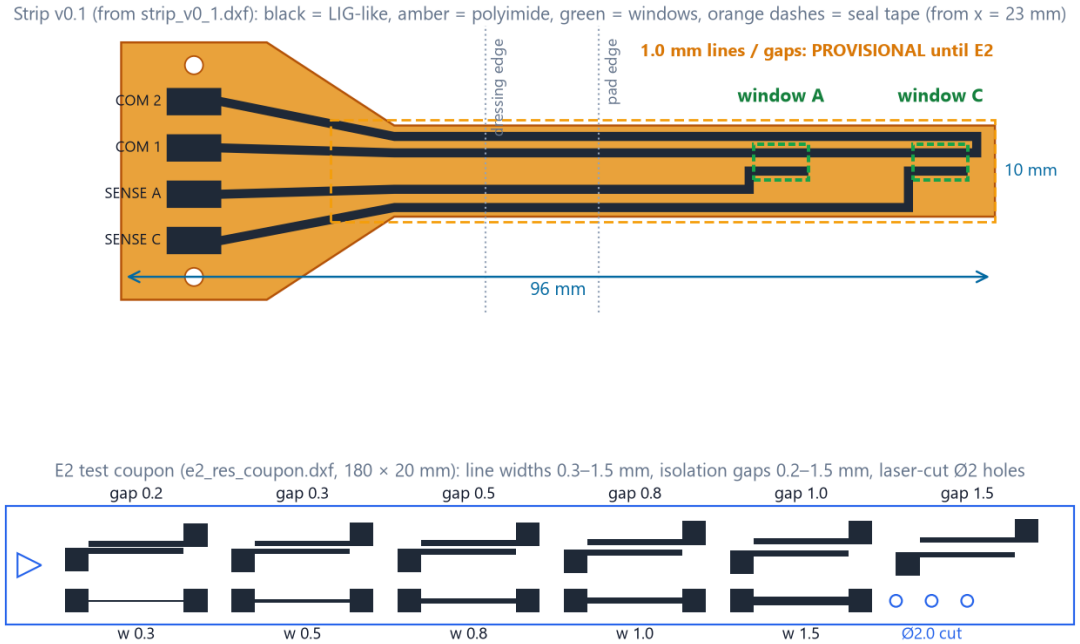


Figure: our laser files drawn from the DXFs: strip v0.1 and the E2 coupon. Not yet lased.

Fabrication Approach & Process Validation (E2)

- **Material:** polyimide written by our CO_2 laser in one raster pass[7]. We call it a LIG-like laser-carbonised conductive layer until Raman confirms graphene.
- **Machine facts (LaserCAD config):** 1200 × 900 mm bed, 20 kHz PWM, 98 % power cap. Our reference '12' is most likely a % PWM command, not watts.
- **Critical parameters logged:** min/max power %, speed, 0.1 mm interval, passes, scan direction, focus and tube current, copied from every layer dialog.
- **E2 plan:** 4 bracket coupons (power 10/12/14, speed ×0.67/1/1.5) on 2 days + 3 resolution coupons; 72 pre-registered samples.
- **Pass rules, fixed before testing:** sheet-resistance CV ≤ 20 %; smallest reliable line and gap; gaps > 3 MΩ; laser-cut holes 2.00–2.10 mm.
- **Sealed leads:** LIG made in air is superhydrophilic[9], so every lead outside the two windows is covered by polyimide tape.

Laser Files Verified: 4 DXFs pass 35/35 checks (mm units, closed outlines, no overlaps, minimum line and gap, fits the bed). The import into LaserCAD is checked at the machine (E2 step S0-6).

Electronics Schematics & BOM

ERC VERIFIED · NOT YET WIRED

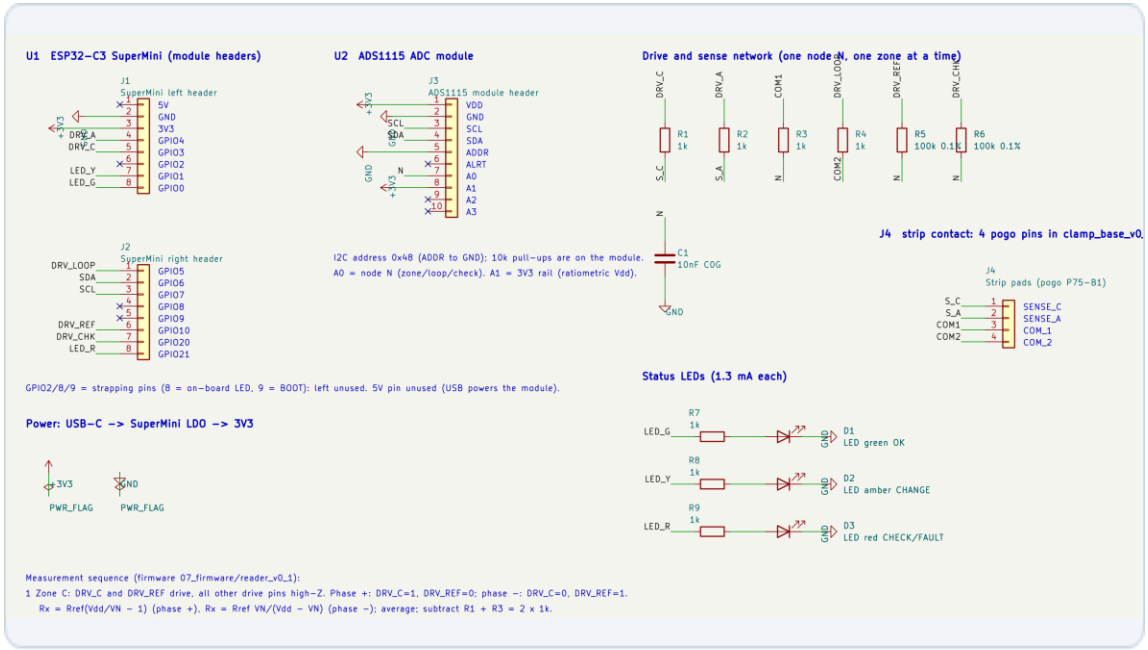


Figure: our KiCad 10 schematic of the two-zone reader, exported by kicad-cli.

- **Power distribution:** USB-C → SuperMini 3V3 regulator → ADS1115 and node network; the 3V3 rail is measured on A1 every scan.
- **Connection strategy:** 4 pogo pins press on the strip's 3 × 6 mm pads (no soldering, no ZIF on porous carbon); 2 dowels through laser-cut holes align it.

Pinout & Connection Map (from the netlist)

Net	ESP32-C3 / module pin	Function
DRV_C · DRV_A	GPIO3 · GPIO4	drive zone C · zone A through 1 kΩ
DRV_LOOP	GPIO5	COM loop continuity (broken strip)
DRV_REF	GPIO10	100 kΩ 0.1 % reference to node N
DRV_CHK	GPIO20	100 kΩ self-check to node N
SDA · SCL	GPIO6 · GPIO7	I²C to ADS1115 (0x48)
N · +3V3	ADS1115 A0 · A1	node voltage · rail (ratiometric)
LED_G/Y/R	GPIO0 · GPIO1 · GPIO21	OK · CHANGE · CHECK/FAULT
S_C S_A COM1 COM2	J4 pins 1–4	pogo pins on the strip pads
not used	GPIO2 · 8 · 9	strapping pins kept free

Verification: KiCad ERC 0 violations. A script checks the netlist against the firmware pins and strip pad order (0 mismatches) and writes the perfboard wiring table.

Electronics Schematics & BOM: Bill of Materials

ESTIMATES · NOT YET ORDERED

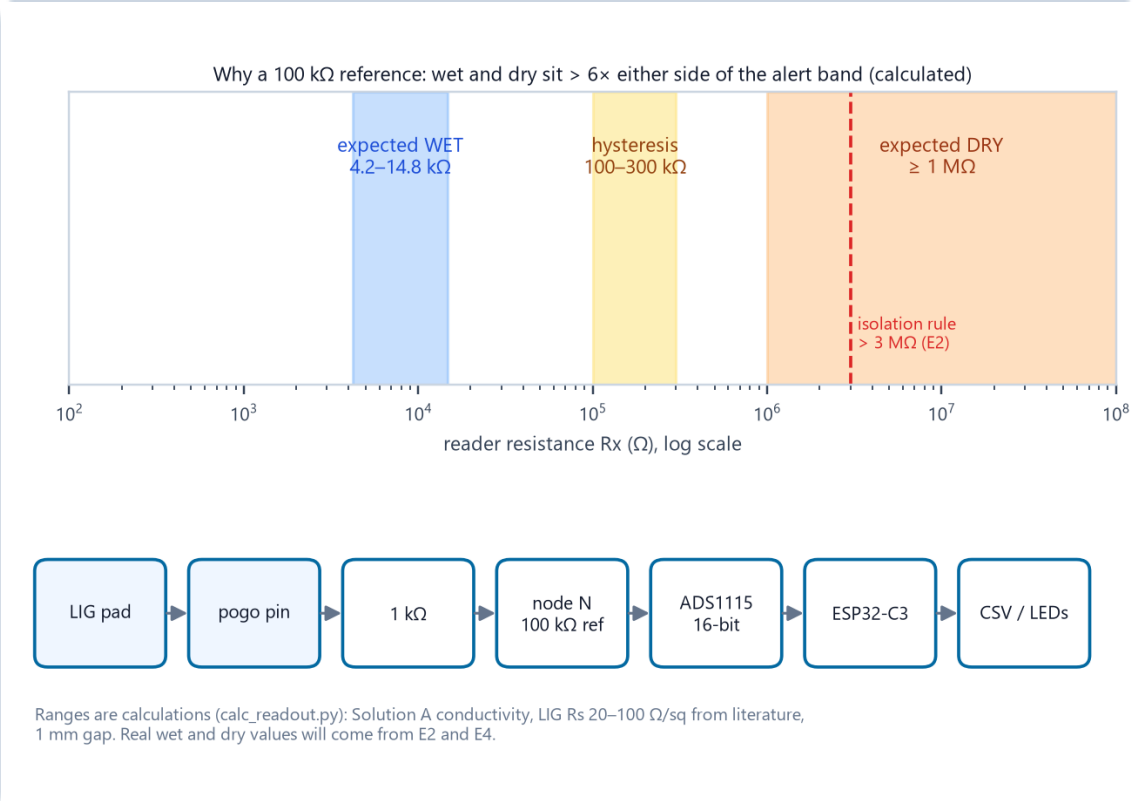


Figure: why these values; ranges are calculations, to be replaced by E2/E4 data.

Itemized Prototype Bill of Materials (estimates)

Component	Specification	Qty	Source	Est. ₹
ESP32-C3 SuperMini	USB-C, native USB serial	1	Local / online	250–450
ADS1115 module	16-bit I²C ADC, auto-range	1	Local / online	200–450
Precision resistors	100 kΩ 0.1 % ×2 · 1 kΩ 1 % ×7	9	Online (element14 / Mouser IN)	47–141
COG capacitor, LEDs, perfboard	10 nF COG · 3 LEDs · headers	set	Local	111–280
Pogo pins P75-B1	1.02 mm, 2 mm travel (4 + 2 spare)	6	Online (Probots)	90–240
Clamp + bench rig	PETG print · acrylic · M3/M4 · dowels	set	Lab / local	100–1,240
Strip materials	polyimide film · seal tape · silver paint	set	Lab / online	800–3,500
Test consumables	1 % resistor set · IPA · gloves · saline	set	Local	480–1,060
Total (excl. DMM, E4 dressings)				**≈ 2,078–7,361**

Procurement Status: nothing ordered yet. Prices are estimates to confirm on supplier pages; all parts are sold in India. First order: silver paint, 1 % resistor set, 0.1 % resistors, ADS1115, ESP32-C3.

Maker Journey

GENUINE ARTIFACTS · NO STAGED EVIDENCE

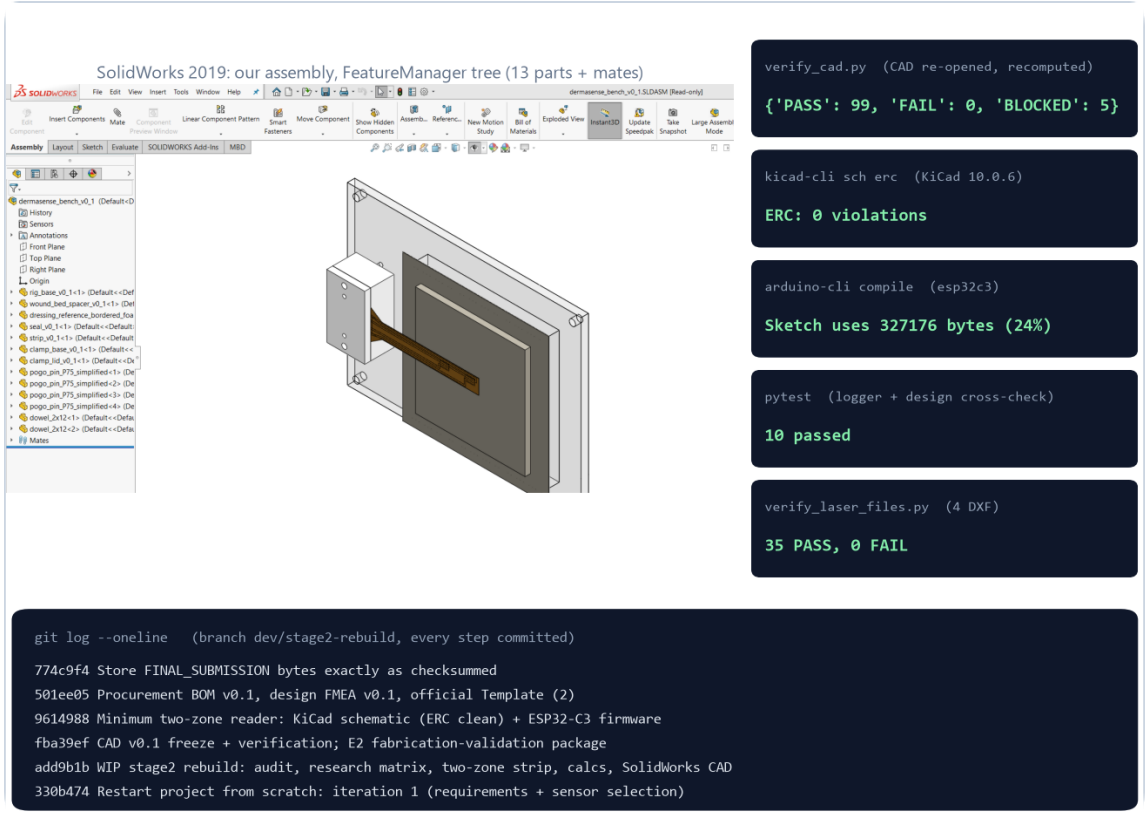


Figure: SolidWorks window captured on our machine + verbatim tool output (14–15 Oct 2026).

Authenticity Verification: What Each Artifact Proves

- **CAD in active use:** our native assembly open in SolidWorks with its FeatureManager tree (13 components + mates); .SLDPRT, .SLDASM, .SLDDRW and .STEP files.
- **Independent re-check:** a script re-opens every CAD file and recomputes volumes and positions from the design data: 99 pass, 0 fail.
- **EDA + firmware:** KiCad schematic with its ERC log (0 violations); firmware built by arduino-cli; logger and cross-check tests pass.
- **Timestamps:** every step is a git commit since the 14 Oct restart, with raw native files (.SLDPRT, .STEP, .kicad_sch, .ino, .dxf) for audit.
- **Lab photographs:** none yet. The first laser session (E2) will be photographed with ID badges. We do not stage or generate lab photos.

Raw Repository Linkage: [team to insert the GitHub / Drive link before upload]: full history from commit 330b474. Faculty sign-off follows the E2 review.

Maker Journey: Engineering Iterations

DOCUMENTED DECISIONS

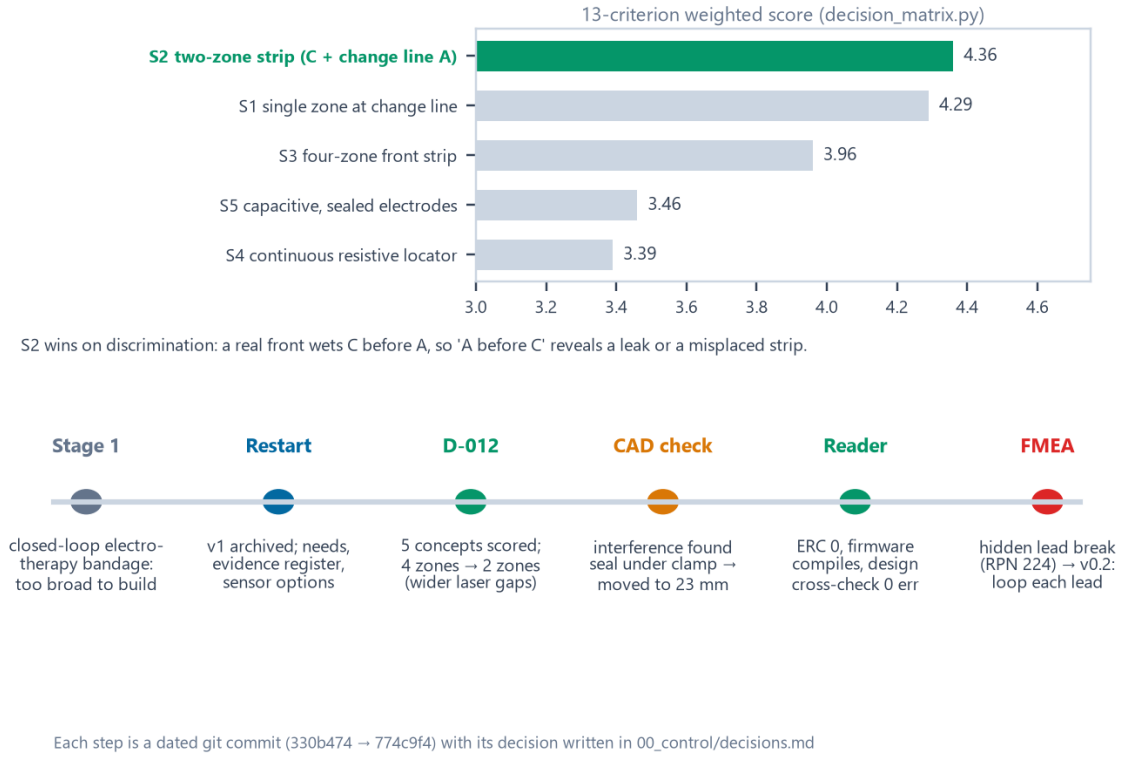


Figure: weighted concept scores and the iteration path; each step is a dated git commit.

Stage 1 to Stage 2 Engineering Iterations

- **Iterative pivot:** Stage 1's closed-loop electrotherapy bandage was too broad to build and validate in four months; we narrowed it to one measurable job: is the dressing full?
- **Concept selection:** 5 sensing concepts scored on 13 weighted criteria. S2 (4.36) beat the simpler S1 (4.29) because its order check separates a real front from a leak.
- **Fit the process we have:** 4 zones → 2 zones, and 1 mm lines and gaps until E2 measures what our laser can resolve.
- **Verification-driven fixes:** seal moved off the clamp; assembly STEP was in cm, now mm; drawings rebuilt first-angle with every dimension once.
- **Risk-driven redesign:** the FMEA showed a cracked sense lead would read 'dry' forever (RPN 224), so v0.2 loops each sense lead to a second pad.

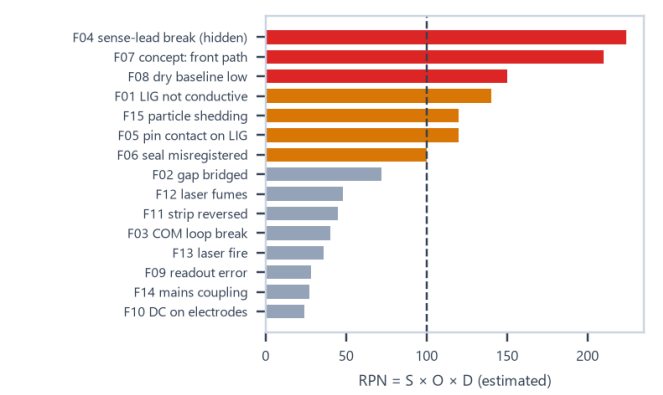
Prior Art We Respect: diaper wetness strips with clip-on readers exist[4]. Our contribution is the change-line placement, the order check and a reproducible laser process.

Safety & Hazard Mitigation

RISKS ESTIMATED · TESTS PLANNED

Design FMEA v0.1: 15 failure modes (ratings estimated before tests)

Failure mode	Effect	RPN	Mitigation / verification
F04 cracked sense lead	zone reads 'dry' forever, no alarm	224	v0.2: loop each lead to a 2nd pad · T-MECH-01
F07 front path differs	CHANGE late or never	210	E4 on real foam dressings
F08 humid dry baseline	false wet	150	measure baselines (E4); thresholds in config.h
F01 weak conductive layer	zones always dry	140	E2 bracket + per-strip QC record
F05 pin contact on porous LIG	readings drift high	120	silver pads, rounded pins · T-CON-01



Physical & Embedded Fail-safes

- **Bench only:** the USB reader is not for patient use; an on-body version must be battery-only and isolated.
- **No net DC:** bipolar drive, high-Z idle; bench test T8 checks < 5 mV average.
- **Self-check every scan:** known 100 kΩ resistor + COM loop → FAULT instead of a wrong state.
- **No wound contact:** strip on the pad's outer face; leads sealed under polyimide tape.
- **Laser work:** signed checklist each session: exhaust, never unattended, CO_2 extinguisher, lowest power first.

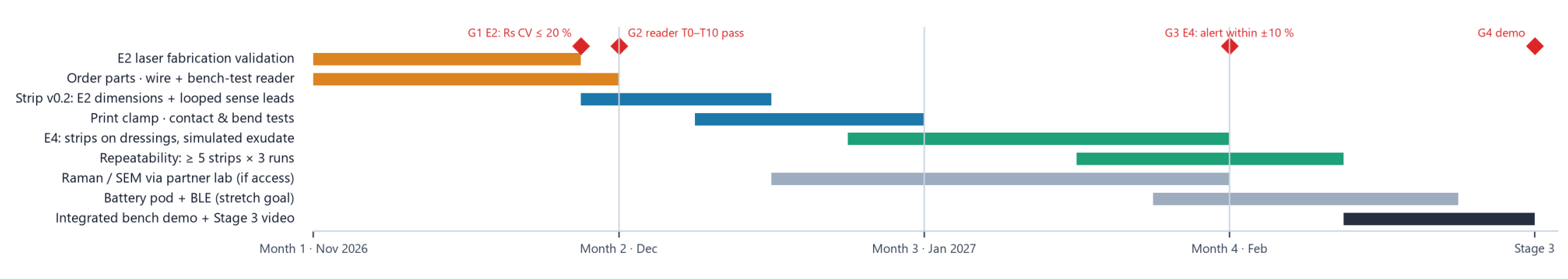
Standards Compliance (considered, not claimed)

- **ISO 14971:** risk-management approach; the FMEA is updated with test data.
- **IEC 60601-1:** electrical safety for any future on-body reader.
- **ISO 10993-1:** biological evaluation before any skin or dressing contact; polyimide film is not sold as medical grade[11].
- **EN 13726:** simulated exudate ('Solution A') for bench dressing tests[10].
- **Laser safety:** IEC 60825-1 class rules and the lab's SOP during fabrication.

Roadmap

FUTURE SCOPE · GATED PLAN

4-Month Fabrication & Validation Plan (each gate must pass before the next step)



Fabrication & Bench Phase (Months 1 - 2)

- **Nov 2026:** E2 laser validation (S0, Day 1, Day 2); order parts; wire and bench-test the reader (T0–T10).
- **Dec 2026:** strip v0.2 from E2 data with looped sense leads; print the clamp; contact (T-CON-01) and bend (T-MECH-01) tests.

Validation & Stage 3 Demo (Months 3 - 4)

- **Jan 2027:** E4: strips on real foam dressings with simulated exudate: dry baselines, thresholds, alert vs the visible front.
- **Feb 2027:** repeatability (≥ 5 strips × 3 runs); integrated bench demo and Stage 3 video with raw data.
- **Stage 3 Target:** a working bench prototype (strip + clamp + reader) showing OK → CHANGE on a real dressing, with logged data. No patient use.

Technical Challenges & Risks

FOUND IN OUR OWN VERIFICATION

Geometry: Seal Trapped Under Clamp

Observed: the 0.09 mm seal tape began at x = 12 mm, but the clamp reaches x = 22 mm: a 9.0 mm³ overlap.

Impact: tape under the paddle lifts the strip off the pogo pins: no contact, or an unreliable one.

How found: automated interference detection in our CAD verification script.

Circuit: A Hidden Sense-Lead Failure

Observed: the reader checks the COM loop, but a cracked SENSE lead simply reads open, which looks like 'dry' (FMEA F04, RPN 224).

Impact: CHANGE would never appear and nothing would alarm.

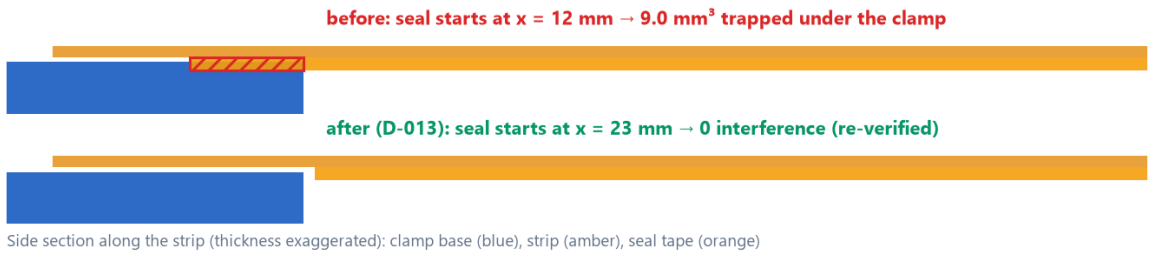
How found: design FMEA of strip + reader together.

Process: An Unknown Laser Recipe

Observed: '12' is a % PWM command on an unidentified tube; CO_2 tubes fire unevenly at low settings[8]. Line width, gap isolation and contact on porous carbon are unmeasured.

Impact: every strip dimension is provisional.

How found: LaserCAD config audit + literature.



Side section along the strip (thickness exaggerated): clamp base (blue), strip (amber), seal tape (orange)



Also Caught in Verification: assembly STEP exported in cm (now mm); drawings showed 0.13 mm for a 0.125 mm film and repeated dimensions (fixed); a SolidWorks API crash on colour arrays (typed arrays).

Potential Solutions & Risk Mitigation

FIXED · DESIGNED · FUTURE SCOPE

Fixed & Verified

Seal: now starts at x = 23 mm; re-verified with 0 interference (D-013).

Units & drawings: all STEP in mm; first-angle A3 drawings; CAD checks 99 pass / 0 fail.

Design consistency: schematic ↔ firmware ↔ strip pads checked by script.

Designed, to Prove on Hardware

v0.2 strip: each sense lead loops back to a second pad (6 pads), so every lead is checked every scan.

Contact: silver-painted pads and rounded pogo tips; contact resistance over clamp cycles (T-CON-01).

Self-test: check resistor and loop → FAULT, verified in bench test T2–T9.

Fallbacks if E2 Disappoints

Wider features: set lines and gaps to what E2 resolves, re-run the DRC.

Process: bracket power and speed, slower or defocused passes, log tube current.

Partner laser: same DXF files on another lab's machine.

Concept: S1 single-zone strip (score 4.29) if two zones cannot be resolved.

Future Scope & Validation Plan: evidence needed before practical or clinical use

- **Remaining physical experiments:** E2 fabrication; reader bench test; E4 wetting tests with simulated exudate on foam dressings; repeatability.
- **Material characterisation:** Raman (D, G, 2D bands) and SEM through a partner facility before any 'graphene' claim.
- **Reliability:** bending (T-MECH-01), contact cycling (T-CON-01), storage and humidity drift.
- **Product steps:** battery-only isolated pod with BLE; multi-strip logging; a strip layout matched to each dressing model's change line.
- **Before any patient use:** ISO 10993 biological evaluation, IEC 60601-1 electrical safety, clinician feedback and an ethics-approved study.
- **Not claimed today:** that the sensor works, that the material is graphene, or any clinical benefit.

Rule We Keep: no claim of sensing performance, graphene identity or clinical benefit until the measurement exists.

References & Sources

SOURCES WITH READING DEPTH

Numbers match the superscripts on each slide. After each source we state how deeply it was read; nothing is labelled verified beyond that.

1. Milne S.D. et al., A wearable wound moisture sensor as an indicator for wound dressing change, Int. Wound J. 13(6):1309–1314 (2016). doi:10.1111/iwj.12521 · full text read

2. Dressing instructions for use: some foam dressings say 'change when exudate approaches ~2 cm from the pad edge'; Mölnlycke Mepilex Border Flex brochure uses strike-through reaching 3 edges · manufacturer documents, read as excerpts

3. McColl D., Cartlidge B., Connolly P., Real-time monitoring of moisture levels in wound dressings in vitro, Int. J. Surg. 5(5):316–322 (2007). doi:10.1016/j.ijsu.2007.02.008 · abstract

4. US 5,469,145 and US 5,838,240 (wet diaper detector: detachable strip + clip-on detector); US 9,291,589; US 10,928,344 · patent summaries, claims not read

5. Jiang Y. et al., Wireless, closed-loop, smart bandage with integrated sensors and stimulators, Nat. Biotechnol. (2022). doi:10.1038/s41587-022-01528-3 · listing + coverage

6. Dealey C., Cameron J., Arrowsmith M., A study comparing two objective methods of quantifying the production of wound exudate, J. Wound Care 15(4):149–153 (2006) · abstract (methods disagree)

7. Lin J. et al., Laser-induced porous graphene films from commercial polymers, Nat. Commun. 5:5714 (2014). doi:10.1038/ncomms6714 · full text read

8. CO_2 laser tube current vs PWM setting, engineering measurements, softsolder.com (2022) · not peer-reviewed

9. Li Y. et al., Laser-induced graphene in controlled atmospheres: from superhydrophilic to superhydrophobic surfaces, Adv. Mater. 29:1700496 (2017). doi:10.1002/adma.201700496 · abstract

10. EN 13726-1:2002 / EN 13726:2023, Test methods for wound dressings (absorbency; 'Solution A') · catalogue entry; online link not reachable from our network

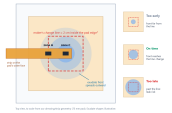
11. DuPont position statement on the use of DuPont materials in medical applications (Kapton caution) · web page

12. Texas Instruments ADS1115 datasheet (SBAS444); Espressif ESP32-C3 datasheet · vendor documents

Image Sources

ALL IMAGES ORIGINAL

Every image in this deck was produced by the team from our own project files. No stock photos, no AI-generated images and no external images are used.



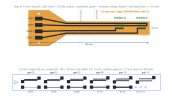
Problem illustration, drawn to scale
Source: make_final_figures.py · strip_v0_1_geometry.json
Stage2_Final/figures/fig_problem.png



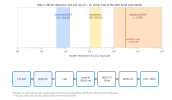
After-state intended workflow
Source: make_final_figures.py · firmware states
Stage2_Final/figures/fig_journey_after.png



Firmware scan and state logic
Source: make_final_figures.py · reader_v0_1 source
Stage2_Final/figures/fig_dataflow.png



Strip and E2 coupon layers
Source: make_final_figures.py · our DXF files
Stage2_Final/figures/fig_strip.png



Measurement window
Source: make_final_figures.py · calc_readout.py
Stage2_Final/figures/fig_window.png



Decision matrix + iterations
Source: make_final_figures.py · decision_matrix.md
Stage2_Final/figures/fig_decision.png



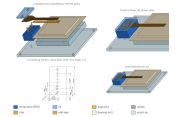
4-month roadmap
Source: make_final_figures.py · E2/bench/E4 plans
Stage2_Final/figures/fig_roadmap.png



Before-state pain-point map
Source: make_final_figures.py · cited sources 1–3
Stage2_Final/figures/fig_journey_before.png



System block diagram
Source: make_final_figures.py · verification logs
Stage2_Final/figures/fig_block.png



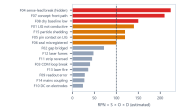
SolidWorks renders + drawing
Source: render_cad.py · 06_cad/solidworks (our models)
Stage2_Final/figures/fig_cad.png



KiCad schematic export
Source: kicad-cli · reader_v0_1.kicad_sch (ours)
Stage2_Final/figures/fig_schematic.png



SolidWorks screenshot + logs
Source: screen capture of our machine · verbatim tool output
Stage2_Final/figures/fig_maker.png



FMEA risk ranking
Source: make_final_figures.py · fmea_v0_1.csv
Stage2_Final/figures/fig_fmea.png



Seal clash + lead-break diagrams
Source: make_final_figures.py · geometry + FMEA
Stage2_Final/figures/fig_challenges.png